HW3 (Chapters 5) for MSE 201:002 Spring 2023

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8^{th} Edition

Chapter 5

Problem 5.7

A sheet of steel 1.5 mm thick has nitrogen atmospheres on both sides at 1200°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is 6×10^{-11} m^2/s, and the diffusion flux is found to be 1.2×10^{-7} $kg/m^2 \cdot s$. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 4 kg/m^3. How far into the sheet from this high-pressure side will the concentration be 2.0 kg/m^3? Assume a linear concentration profile.

Problem 5.8

A sheet of BCC iron 1 mm thick was exposed to a carburizing gas atmosphere on one side and a decarburizing atmosphere on the other side at 725°C. After reaching steady state, the iron was quickly cooled to room temperature. The carbon concentrations at the two surfaces of the sheet were determined to be 0.012 and 0.0075 wt%. Compute the diffusion coefficient if the diffusion flux is 1.4×10^{-8} $kg/m^2 \cdot s$. *Hint:* Use Equation 4.9 to convert the concentrations from weight percent to kilograms of carbon per cubic meter of iron.

Problem 5.22

The diffusion coefficients for silver in copper are given at two temperatures:

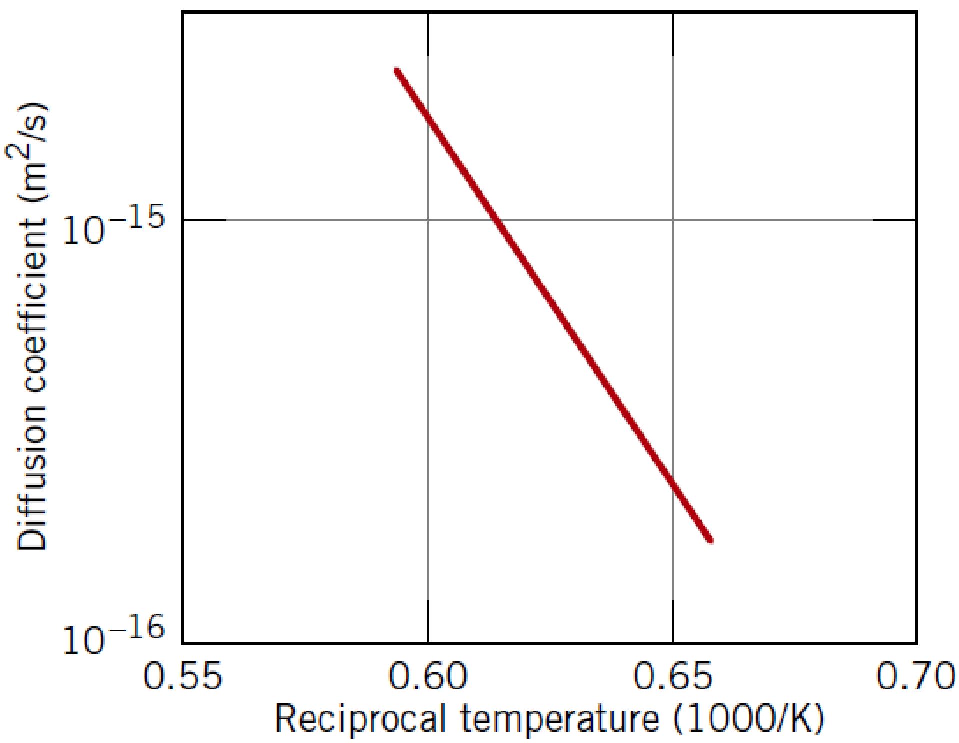
T(°C)	*D*(m^2/s)
650	5.5×10^{-16}
900	1.3×10^{-13}

(a) Determine the values of D_0 and Q_d.

(b) What is the magnitude of D at 875°C?

Problem 5.23

The following figure shows a plot of the logarithm (to the base 10) of the diffusion coefficient versus reciprocal of the absolute temperature, for the diffusion of iron in chromium. Determine values for the activation energy and preexponential.



Diffusion coefficient (m2/s)
10−15
10−16
0.55
0.60
0.65
0.70
Reciprocal temperature (1000/K)

Problem 5.D2

5.D2 A gas mixture is found to contain two diatomic A and B species for which the partial pressures of both are 0.05065 MPa (0.5 atm). This mixture is to be enriched in the partial pressure of the A species by passing both gases through a thin sheet of some metal at an elevated temperature. The resulting enriched mixture is to have a partial pressure of 0.02026 MPa (0.2 atm) for gas A, and 0.01013 MPa (0.1 atm) for gas B. The concentrations of A and B (C_A and C_B, in mol/m^3) are functions of gas partial pressures (p_{A_2} and p_{B_2}, in MPa) and absolute temperature according to the following expressions:

$$C_A = 200\sqrt{p_{A_2}}\exp\left(-\frac{25.0 \text{ kJ/mol}}{RT}\right) \quad (5.18a)$$

$$C_B = 1.0 \times 10^3\sqrt{p_{B_2}}\exp\left(-\frac{30.0 \text{ kJ/mol}}{RT}\right) \quad (5.18b)$$

Furthermore, the diffusion coefficients for the diffusion of these gases in the metal are functions of the absolute temperature as follows:

$$D_A(\text{m}^2/\text{s}) = 4.0 \times 10^{-7}\exp\left(-\frac{15.0 \text{ kJ/mol}}{RT}\right) \quad (5.19a)$$

$$D_B(\text{m}^2/\text{s}) = 2.5 \times 10^{-6}\exp\left(-\frac{24.0 \text{ kJ/mol}}{RT}\right) \quad (5.19b)$$

Is it possible to purify the A gas in this manner? If so, specify a temperature at which the process may be carried out, and also the thickness of metal sheet that would be required. If this procedure is not possible, then state the reason(s) why.